RV Wall Panel & Smart Awning Controller

An open DIY build for Home Assistant using two ESP32 boards running ESPHome. Version 1.3, October 2026.

Wall panel controller (inside)

- Battery voltage, current, power and state of charge (INA226 + shunt)
- Fresh, grey and black tank levels
- Water pump and gas water heater on/off from Home Assistant
- Water main shutoff and tank dump valves; Home Assistant closes the main if a wireless leak probe gets wet
- Four silent solid-state relays. Pump, heater and dump valve always start OFF after a reboot; a leak shutoff stays shut

Smart awning controller (outside)

- One box on the outer arm, next to the motor: motor driver, current sensor and accelerometer together
- Live awning position 0-100% from the accelerometer
- Detects a jam, a stall or a slow opening, and stops the motor
- Stops if it goes past full open: this motor doesn't stall at the end, it keeps going and rolls the fabric back up upside down. The arm tilt catches that, with the reed switch as a backup
- Feels the awning shaking in wind and retracts on its own, even with Home Assistant or Wi-Fi down (optional)
- The factory wall rocker still works as a backup, even if the ESP32 is dead
- Reed switch confirms "fully closed"

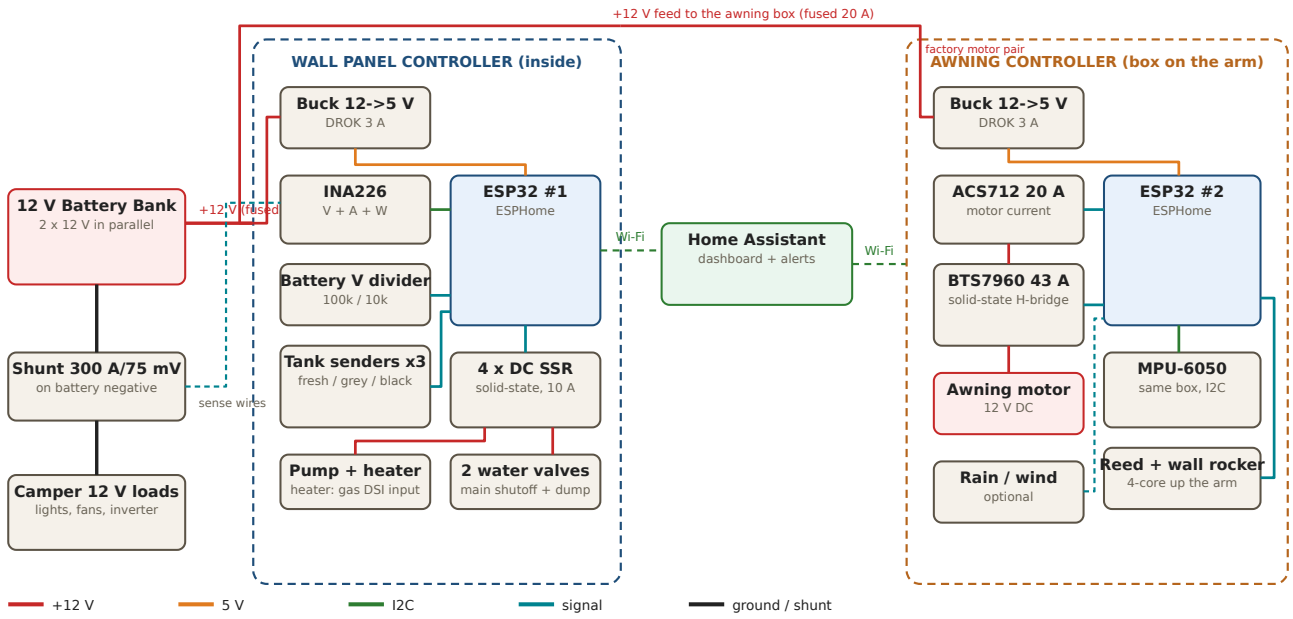


Figure 1: System overview. Both controllers report to Home Assistant over Wi-Fi. The awning box runs on its own: position stops, stall checks and wind retract work without Home Assistant.

Project status: the firmware for both controllers compiles (ESPHome 2026.10). The wall panel firmware has been flashed and tested on a bench board with nothing wired to it. The awning hardware has not been built or field-tested yet. Thresholds such as stall current, wind shake and the tilt-to-position points are starting values, so tune them on your own rig.

1. Parts list

The listings are examples that were available in October 2026. Any part with the same spec works.

All boards

Part	Spec	Qty	Used for	Example listing (Amazon ASIN)
ESP32 DevKit (ESP32-WROOM-32)	esp32dev board, 30/38 pin	2	Wall panel, awning box	any ESP32 DevKit V1
12 V to 5 V buck converter	8-35 V in, 5 V 3 A out, potted/waterproof	2	Powers each ESP32 (and the ACS712, GY-521)	DROK B07P663XJV
Inline ATC fuse holders + fuses	12 AWG holders; 1 A, 3 A, 10 A, 20 A fuses	1 kit	Fuse every feed at the battery end	DaierTek 12 AWG holders B09CYRQ46S + ZIPCCI fuses B0B18NVT5Z
Hook-up wire	22 AWG (signals), 14 AWG (pump/motor)	-	Wiring	-
Resistors, 1/4 W 1% metal film	10k (x9), 100k (x1), tank R ref (see notes)	-	Dividers, SSR/driver input pull-downs	EDGELEC 10k B07HDGX5LM
Ceramic capacitors 100 nF	50 V	~6	ADC noise filtering	any

Wall panel controller (inside)

Part	Spec	Qty	Used for	Example listing (Amazon ASIN)
INA226 current/voltage board	I2C, 0-36 V bus; desolder onboard R100	1	Battery volts, amps, watts	Qoroos 3-pack B0FLD93VRC
Current shunt	300 A / 75 mV (0.00025 ohm)	1	Battery negative line	Fielect FL-2 B082GTY6YX
DC solid-state relay (SSR)	DC-to-DC MOSFET; input 3-32 V (ESP32 3.3 V pin drives it directly); output 5-30 V, 10 A (20 A < 2 s); sealed, wire leads	4	Water heater, water pump, water main valve, dump valve	OONO F-1049 B09T3RT5MG
Motorized water valves, 12 V, 3-wire	power + ground + control wire. Main: 12 V on control = CLOSED. Dump: 12 V on control = OPEN	2	Water main shutoff (city inlet) + tank dump	match your pipe size
Water leak probes	wireless Zigbee (e.g. Aqara), joined to Home Assistant	1 per spot	Leak shutoff: HA tells the panel to close the main	Aqara water leak sensor
Tank level senders	resistive senders (e.g. 240-33 ohm or 0-190 ohm)	3	Fresh / grey / black	match your tanks
Enclosure	about 170 x 100 mm and 65-70 mm deep inside (see Figure 5)	1	Holds both decks, inside a cabinet	Zulkit 200 x 120 x 75 (same as the awning

				box) B07RT6NWTR
Standoffs + deck plate	M3 nylon standoffs 30 mm (x4) and screws; 2-3 mm ABS/acrylic plate or a large perfboard, about 160 x 90 mm	1 set	Logic deck above the relays	any M3 nylon standoff kit

Awning controller (outside)

Part	Spec	Qty	Used for	Example listing (Amazon ASIN)
MPU-6050 accelerometer (GY-521)	I2C, 3.3 V	1 (+spares)	Arm angle -> position + wind	HiLetgo 3-pack B00LP25V1A
IP65 box	200 x 120 x 75 mm	1	The awning box on the outer arm: everything	Zulkit B07RT6NWTR
PG7 cable glands	3-6.5 mm, nylon IP68	pack	Sealed cable entries	MAKERELE B0CTCHB115
Shielded cable	22 AWG 4-conductor, flexible	~4 m	Reed switch + wall rocker, up the arm next to the factory motor pair	UL2464 22/4 B0D9JY3MC4
Magnetic reed switch + magnet	N.O., sealed, wired	1	'Awning closed' sensor	NO/NC waterproof B0F2F99Q2C
ACS712 current sensor, 20 A	5 V module, 100 mV/A	1	Motor stall/jam detection	3-pack B08R5RBNFK
Motor driver, BTS7960 (IBT-2)	43 A solid-state H-bridge, 6-27 V; VCC on 3.3 V	1	Runs the awning motor both ways, brakes when off	HiLetgo B00WSN98DC
Manual-override parts	2 x 1N5817 Schottky diodes, 2 x 33k resistors	-	Keeps the wall rocker as a backup switch	any
Cup anemometer	reed-switch pulses, 1 Hz = 1.492 mph; RJ11 plug (use the 2 middle wires)	1 (optional)	Wind speed in Home Assistant (alerts, history)	MISOL wind speed sensor B00QDMBQGG
Rain sensor	switch output	optional	Extra weather input	-

Relay sizing: skip the cheap multi-channel "solid state relay boards" (Omron G3MB-202P and copies). They are AC-only: on 12 V DC they switch on and then stay stuck on, and 2 A is far too small for a water pump (5-7 A) anyway. For the wall panel, use a **DC-to-DC** MOSFET solid-state relay rated at least 10 A at 5-30 V whose input turns on from 3 V, so the ESP32 can drive it directly. For the awning motor, use a BTS7960 H-bridge board: it reverses the motor with no contacts to wear, brakes it when off, and shuts itself down on over-current or over-heat. Avoid "40 A DC SSR" blocks (Fotek SSR-40DD and clones): most are counterfeit, run hot, usually fail stuck ON, and can't reverse a motor.

2. Wall panel controller wiring

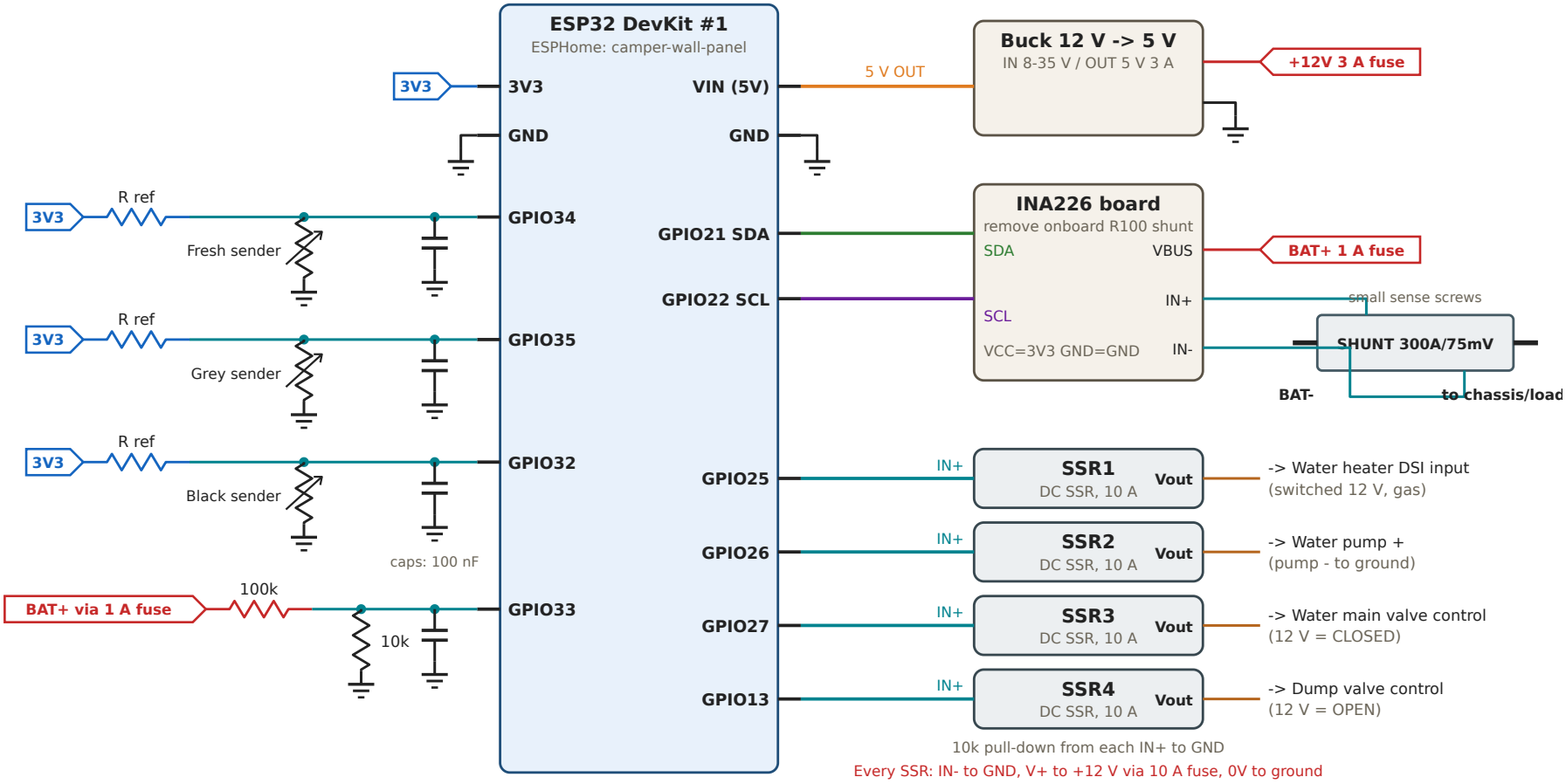


Figure 2: Wall panel schematic. Pins are grouped by function, not in their physical order on the board. Net flags (3V3, +12V, BAT+) connect to the matching supply.

Wall panel pin map

ESP32 pin	Function	Notes
GPIO34	Fresh tank sender (ADC)	3V3 - R ref - node - sender - GND
GPIO35	Grey tank sender (ADC)	same
GPIO32	Black tank sender (ADC)	same
GPIO33	Battery voltage (ADC)	BAT+ - 1 A fuse - 100k - node - 10k - GND
GPIO21 / 22	I2C SDA / SCL	INA226 at 0x40
GPIO25	SSR1: water heater (gas)	Always OFF at boot
GPIO26	SSR2: water pump	Always OFF at boot; blocked during a leak lockout
GPIO27	SSR3: water main valve	ON = closed; kept through a reboot
GPIO13	SSR4: tank dump valve	ON = open; OFF at boot, auto-closes after 15 min

- **INA226 with an external shunt:** desolder the small `R100` resistor on the board. Run IN+ and IN- to the two small sense screws on the 300 A shunt. VBUS goes to battery + through a 1 A fuse. If charging reads negative, set `current_sign: "-1.0"`.
- **Battery voltage divider:** 100k over 10k gives a ratio of 11. Measure with a multimeter and adjust `battery_divider_ratio` until the reading matches.
- **Tank senders:** choose R ref to suit the sender: about 150 ohm for 240-33 ohm senders, or 220 ohm for 0-190 ohm senders. Then enter the measured empty and full voltages in the `*_v_empty / *_v_full` substitutions. Many RV tanks use probe sensors rather than resistive senders. Check yours before wiring.
- **Ground:** take the ESP32/buck ground from the *load side* of the shunt (chassis), like every other load, so its own current is counted.
- **Solid-state relays:** each one has six leads. **IN+** goes to its ESP32 pin, with a 10k pull-down to GND so it stays off while the ESP32 boots. **IN-** goes to the ESP32 ground. On the output side, **V+** goes to +12 V (one shared 10 A fuse), **0V** to ground, and **Vout** is the switched +12 V to the load. The second 0V lead can carry the load's return or be capped. They are sealed with screw tabs and screw to the box floor in two rows of two (the power deck); the ESP32 and the small boards sit on a plate above them (Figure 5). No jumpers and no 5 V supply needed.
- **Water valves:** use 12 V motorized valves with **three wires** (+12 V, ground, and a control wire). Their +12 V and ground stay connected; the SSR switches only the control wire. A solid-state relay has no "normally closed" side, so buy the main shutoff as the type where **12 V on the control wire = CLOSED**. That way the water stays on whenever the controller is off or rebooting, like a normal house. Buy the dump valve as the type where **12 V on control = OPEN**; the firmware closes it at boot and after 15 minutes. A 2-wire valve that reverses polarity (for example Valterra Drain Master) can't be run from one SSR.
- **Battery size:** set `battery_capacity_ah` to your whole bank, for example 200 for two 100 Ah batteries in parallel. The shunt values in the firmware already match the 300 A / 75 mV shunt.
- **Water heater:** the SSR's Vout feeds the same 12 V signal the factory wall switch sends to the DSI board. It can be wired in parallel with the factory switch, so either one turns the heater on. Leave all of the appliance's own safety controls in place.

3. Awning controller wiring

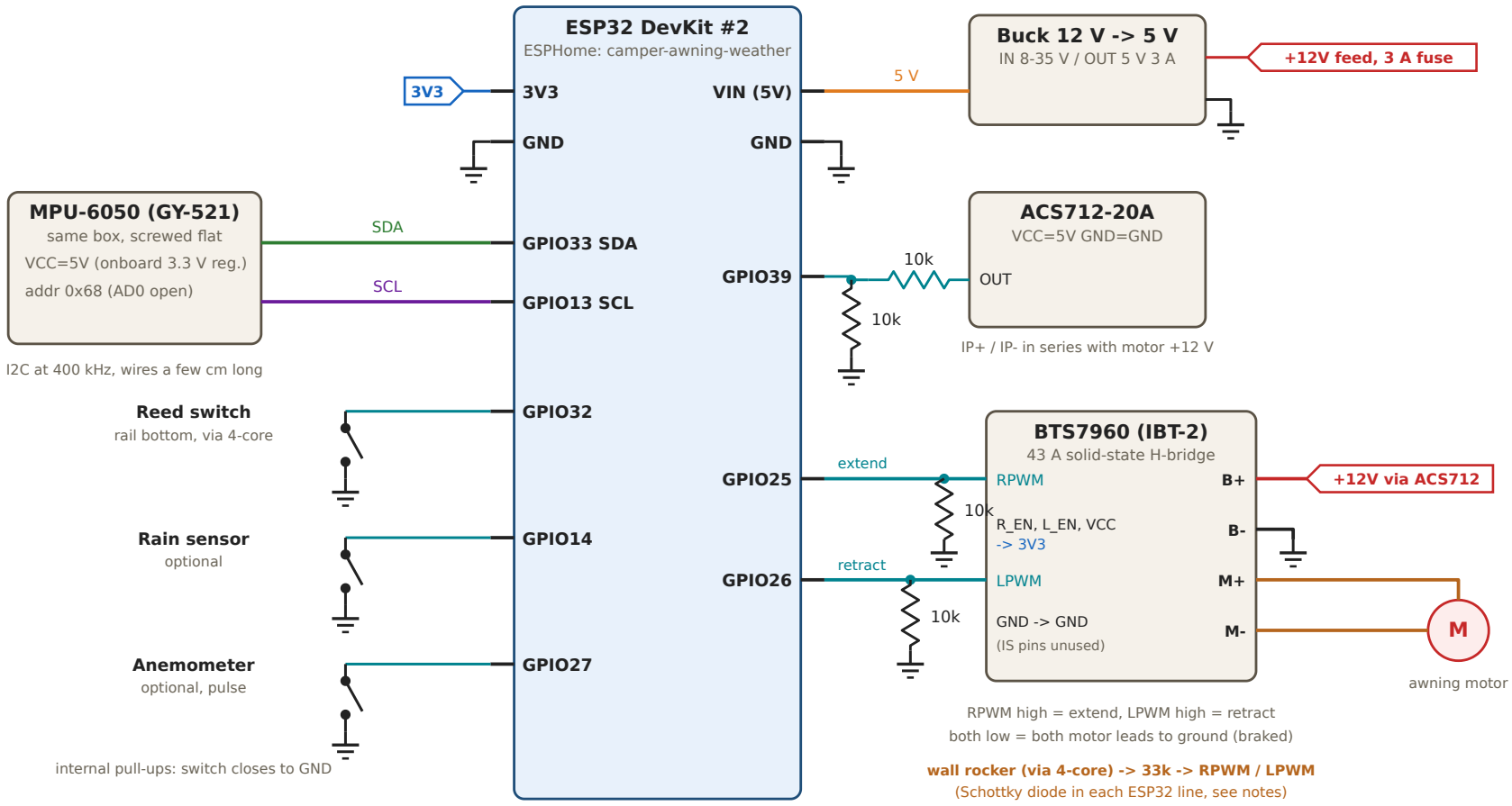


Figure 3: Awning controller schematic (everything in the box on the arm). A BTS7960 board is a solid-state reversing H-bridge. With both inputs low, both motor leads sit at ground, so the motor is stopped and braked.

Awning pin map

ESP32 pin	Function	Notes
GPIO33 / 13	I2C SDA / SCL (400 kHz)	MPU-6050 at 0x68, same box
GPIO32	Reed switch 'closed'	at the bottom of the rail, via the 4-core; to GND, internal pull-up
GPIO39	ACS712 OUT via 10k/10k divider	zero point learned automatically
GPIO25	BTS7960 RPWM: extend	interlocked with retract
GPIO26	BTS7960 LPWM: retract	interlocked with extend
GPIO14	Rain sensor (optional)	to GND
GPIO27	Anemometer pulses (optional)	to GND

- **Motor driver:** B+ to +12 V through the ACS712 (20 A fuse at the battery), B- to ground, M+ and M- to the two motor leads. GPIO25 to RPWM (extend), GPIO26 to LPWM (retract), each with a 10k pull-down to GND so the motor can't move while the ESP32 boots. The firmware interlocks them and waits 500 ms between directions. If the awning runs the wrong way, swap M+ and M-.
- **ACS712 output:** it swings up to 5 V, so always use the 10k/10k divider before GPIO39. That gives 20 A per volt at the pin (`motor_amps_per_volt: "20.0"`).
- **Power feed:** the factory motor pair already runs from the roller tube down the arm to the wall. Disconnect it from the rocker at the wall and feed it from the + bus (20 A fuse near the battery): it becomes the box's 12 V feed. Inside the box, a 3 A inline fuse protects the buck lead. The motor connects to the BTS7960 with short new leads.
- **MPU-6050:** GY-521 VCC on 5 V (it has its own 3.3 V regulator, so SDA/SCL stay at 3.3 V), SDA to GPIO33, SCL to GPIO13, AD0 and INT unconnected. Screw it down flat and solid: if it can wobble, every bump looks like wind.
- **Driver logic power:** connect the BTS7960's VCC, R_EN and L_EN to the ESP32's **3V3** pin, not 5 V. Its input buffer then reads the ESP32's 3.3 V signals reliably. Leave R_IS and L_IS unconnected; the ACS712 does the current sensing. Mount the board heatsink-up with some air around it. An awning motor only draws a few amps, so it stays cool.

> **Keep the wall rocker as a backup switch.** The factory rocker can't stay on the motor (the driver holds both motor leads at ground, so pressing it would short +12 V), and its old pair is now the power feed. Re-wire it at the wall instead: rocker common to +12 V through a 1 A fuse, and its "extend" and "retract" contacts onto two wires of the 4-core cable up the arm. In the box, each goes through a 33k resistor to RPWM / LPWM. Put a 1N5817 Schottky diode in each ESP32 line (stripe toward the driver) so the rocker and the ESP32 can't fight. The 33k and the 10k pull-down bring 12-14 V down to about 3 V. The rocker then drives the motor directly, even if the ESP32 has crashed or isn't flashed. A rocker run bypasses the firmware's jam, stall and past-full-open checks (let go at full open, as with the factory switch), but the position is still tracked, and the wind check pauses while the motor runs. If the rocker and the ESP32 ever ask for opposite directions at once, the driver brakes the motor; it can't short.

4. Box layouts (parts placement)

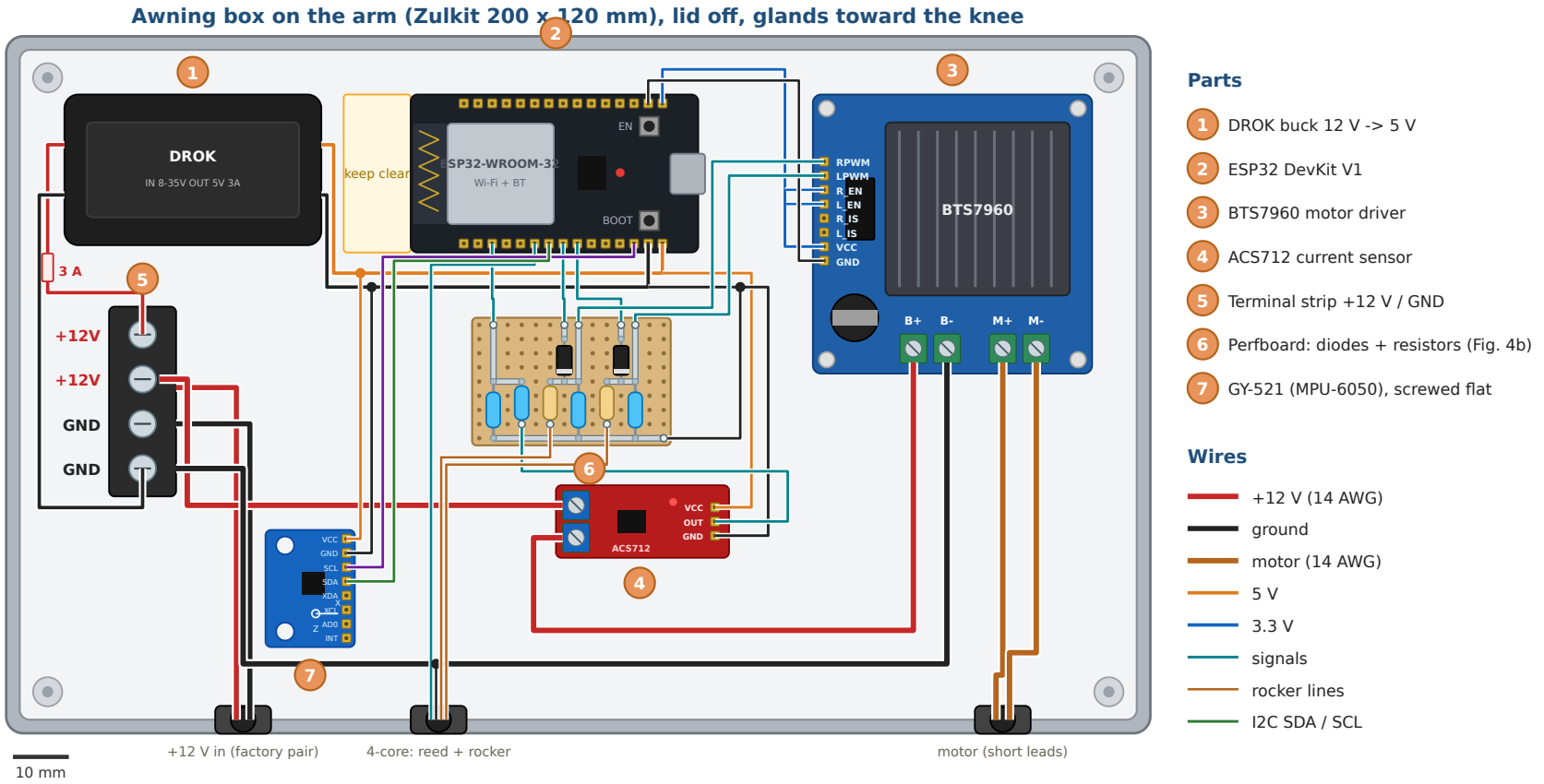


Figure 4: The awning box, drawn to scale from the real parts' sizes. It mounts on the outer arm by the roller tube. Glands go on the edge facing the knee, pointing down when the awning is open, so water can't run in. The BTS7960 sits heatsink-up on the right with its power terminals toward the short motor leads. Your boards may differ slightly in color or pin order, so always follow the labels printed on them.

- Mount each board on 6-10 mm nylon standoffs (M3), or on a plastic back plate screwed to the box's bosses. Nothing metal should touch the bottom of a board.
- Keep the ESP32's antenna end (the end without the USB socket) clear of wires and metal, and away from the heatsink, for the best Wi-Fi.
- Use 14 AWG for the +12 V, ground and motor runs (the thick lines). 22 AWG Dupont jumpers or solid wire are fine for the signals.
- Crimp ferrules on stranded wire going into screw terminals, and add a small drip loop below every gland outside the box.
- This box is big for an arm: check it clears the rail, the inner arm and the gas strut when the awning folds closed before drilling anything.

4b. Awning perfboard close-up

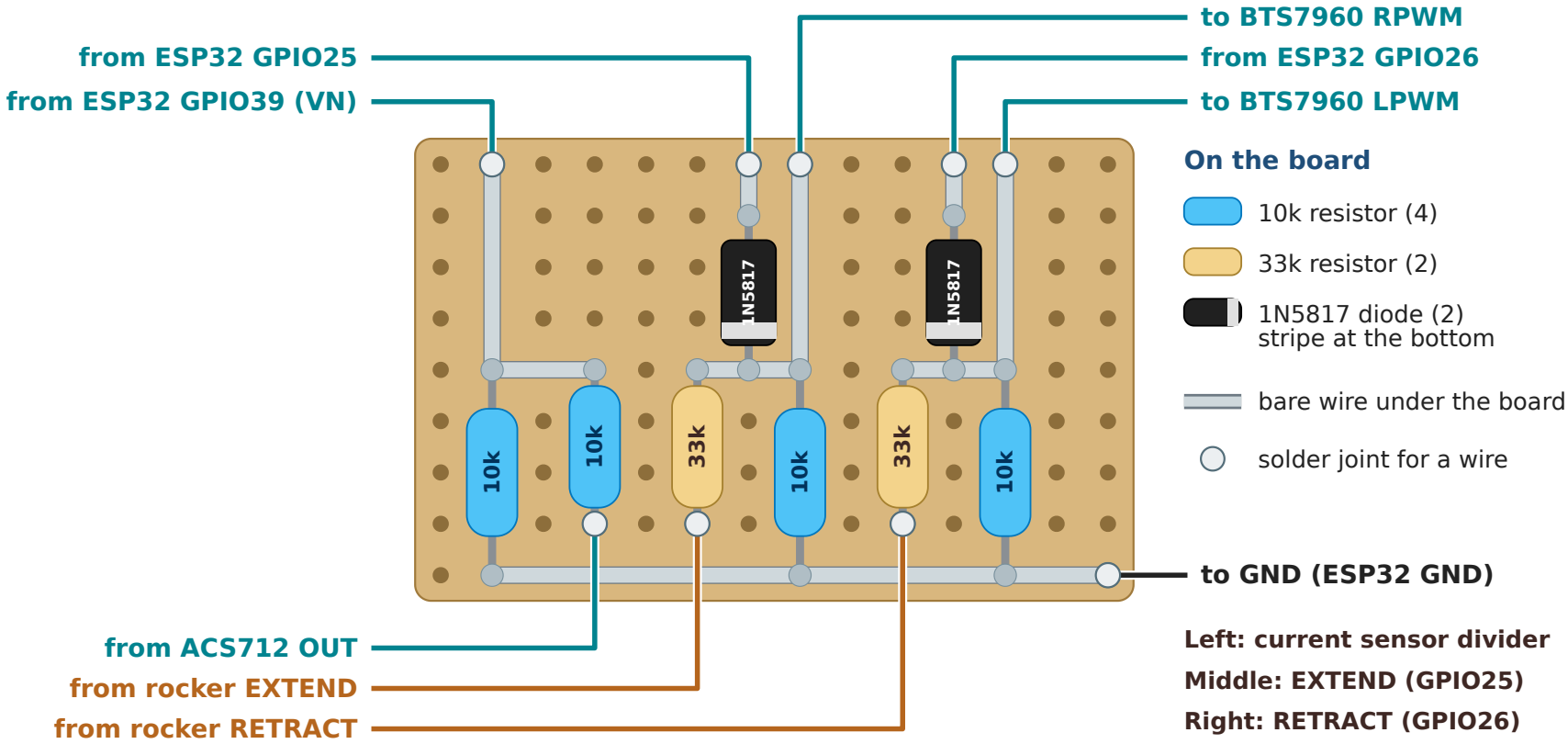


Figure 4b: The awning perfboard from the top, drawn about 3.5x size (14 x 9 holes, about 36 x 23 mm). Each part's value is printed on it; grey lines are bare wire on the underside.

1. Cut a 14 x 9 hole piece. Fit the 8 parts on the holes shown. Both **diodes** go **stripe at the bottom** (toward RPWM / LPWM); backwards, the ESP32 can't run the motor.
2. Underneath, lay bare wire (a cut-off resistor lead works) along each grey line and solder every lead it touches. The long bottom wire is ground.
3. Solder the off-board wires on the white dots. The ACS712 and rocker wires cross the ground wire, so keep them insulated there and solder each to its resistor's lead.
4. Check with a meter first: GPIO25 to RPWM reads as a diode (about 0.2 V one way, open the other). RPWM and LPWM read 10k to GND, the rocker wires 43k, GPIO39 5k.

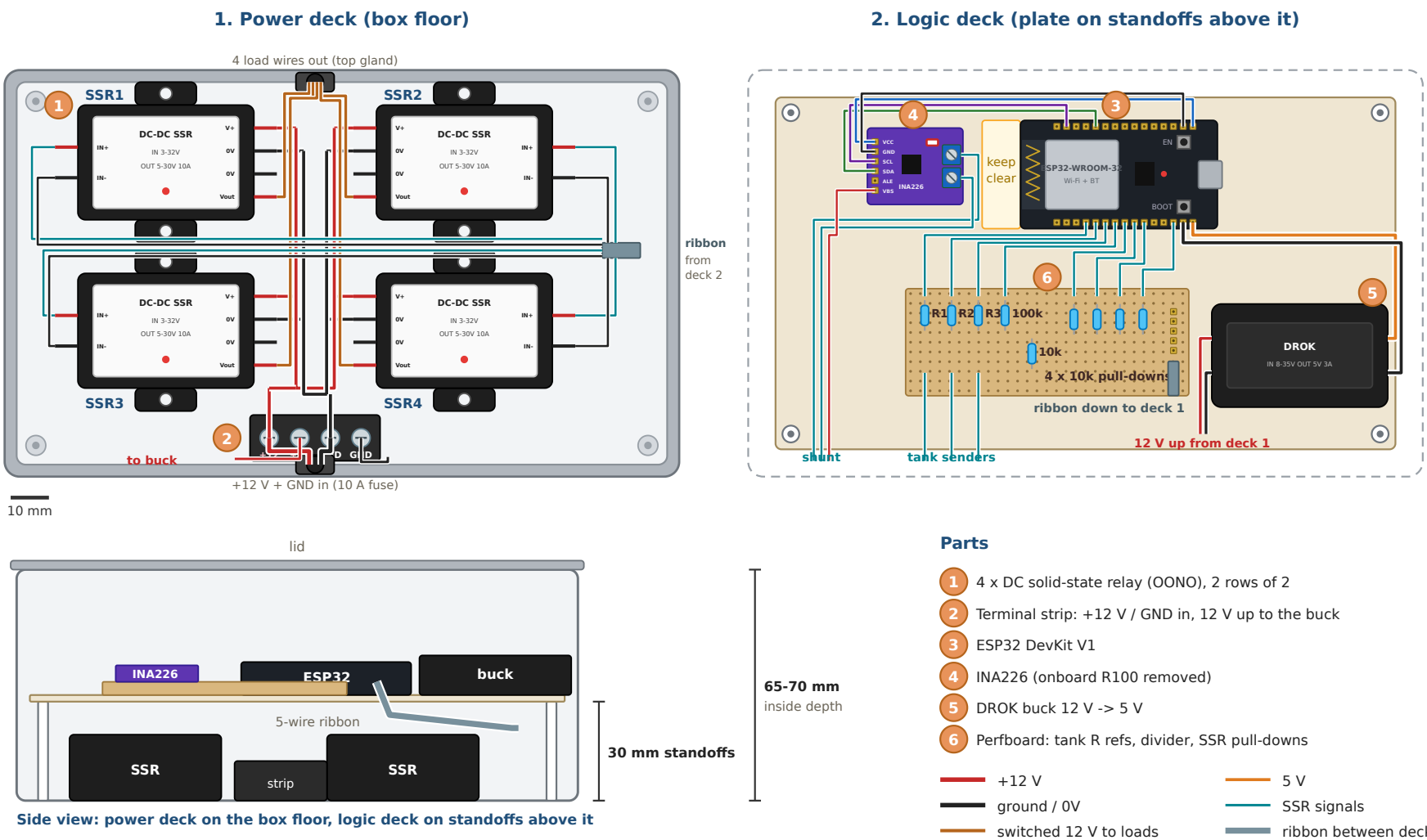


Figure 5: Wall panel box, drawn to scale (the relays' size is approximate, so check yours). Everything is stacked in two decks. The power deck on the box floor holds the four solid-state relays and the terminal strip. The logic deck is a plate on four 30 mm standoffs above it, holding the ESP32, buck, INA226 and perfboard. A 5-wire ribbon (4 signals + ground) and a 2-wire 12 V lead join the decks. The 300 A shunt mounts by the battery; only its thin sense wires come in.

5. Whole-system power wiring

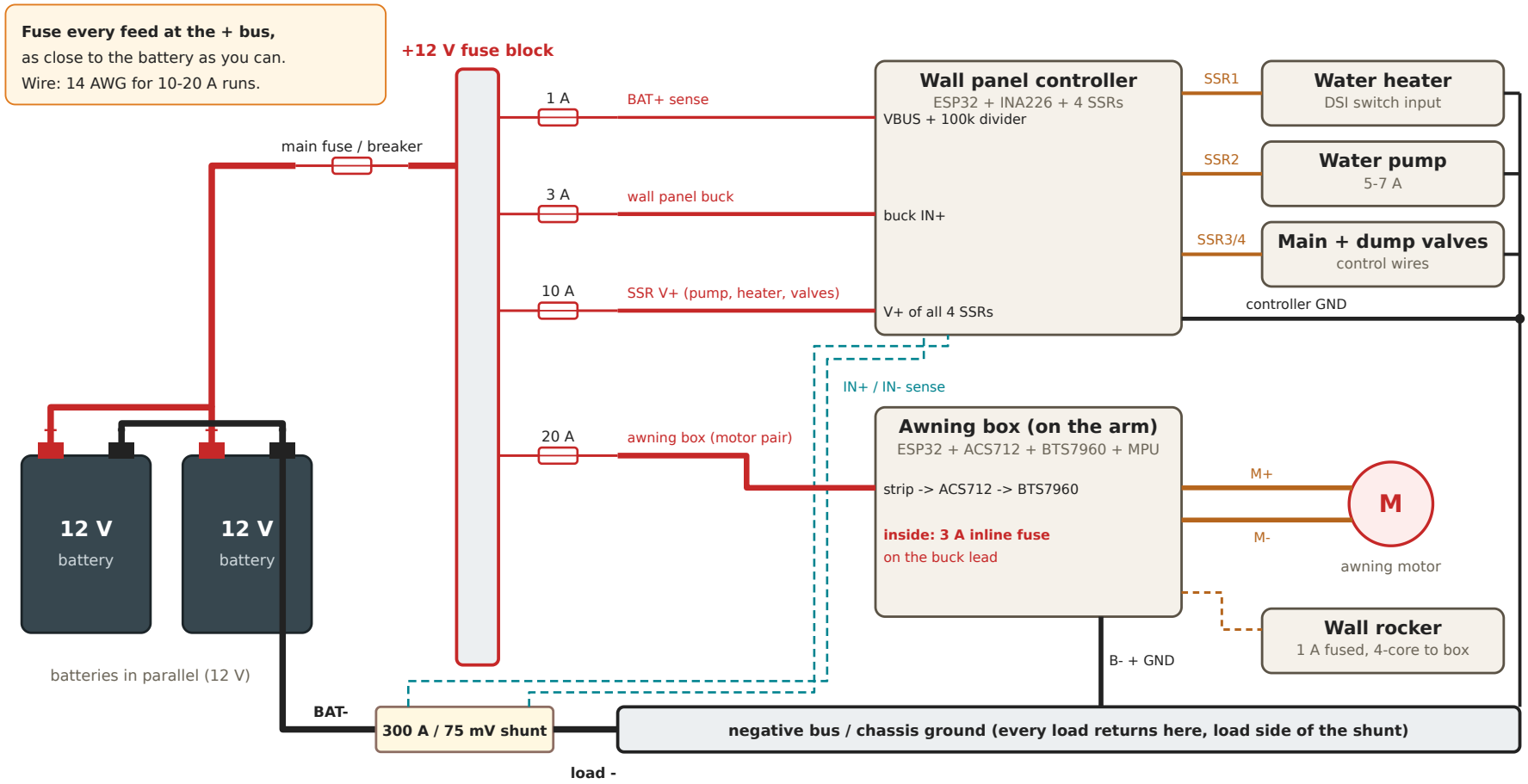


Figure 6: Power distribution. Every feed is fused at the + bus. The shunt sits between the battery negative and everything else, so the INA226 sees all of the current going in and out of the battery.

6. Mounting the awning sensors (Lippert Solera arm)

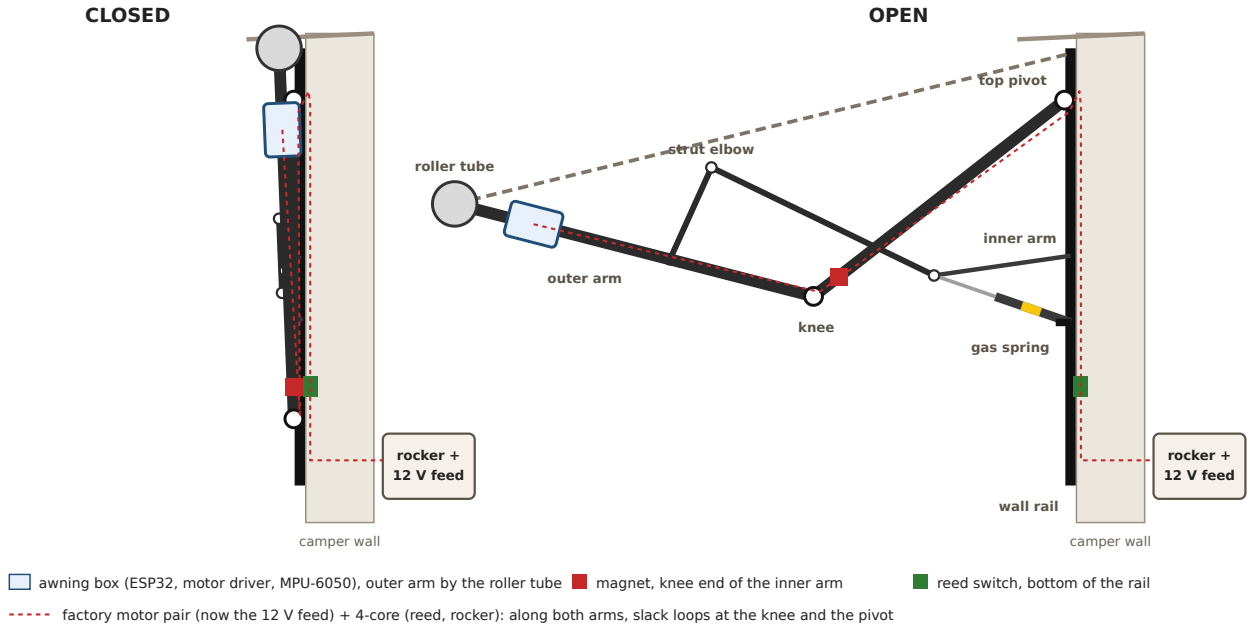


Figure 7: Side view of a Lippert Solera-style 12 V power awning arm. The inner arm pivots at the top of the wall rail and the outer arm runs from the knee to the roller tube, with a strut (a rod plus a gas spring from the rail, out to an elbow, then a link down to the outer arm). Closed, everything folds flat into the rail with the knee at the bottom. The awning box rides on the outer arm near the roller tube, next to the motor: standing straight up when closed, close to flat when open.

- Mount the awning box **flat on the outer arm, close to the roller tube**, near where the motor wires come out. The outer arm swings from vertical (closed) to nearly horizontal (open), so the box's tilt gives the position. Out at the tip, it also feels the most wind shake.
- The arm's hinges make that tilt change unevenly as the awning opens, so the firmware maps it to 0-100% from a few measured points (see section 7).
- Check the box clears the rail, the inner arm and the gas strut when the awning folds closed. If it doesn't, move it along the arm or onto the arm's side face (the board must still stand up when closed and lie flat when open).
- Mount the magnet on the **knee end of the inner arm** and the reed switch on the **bottom of the wall rail**, where they meet when the awning is fully closed.
- The factory motor pair already runs down the arm: it becomes the 12 V feed. Run the 4-core cable beside it (along the outer arm, past the knee, up the inner arm, over the top pivot, down the rail, about 4 m) to the reed switch and the wall rocker, with slack loops at the knee and the pivot.
- Use stainless screws or VHB tape, and don't drill into the arm's spring channel. Seal every cable entry with a PG7 gland.

> **Why an accelerometer?** A single MPU-6050 replaces a stack of tilt or limit switches. It gives continuous position (so the firmware can spot a slow or stuck run anywhere along the travel), and it doubles as a wind sensor, the same idea commercial awning wind sensors use.

7. Firmware and setup

Both controllers run **ESPHome**. The YAML files are `camper-wall-panel.yaml` and `camper-awning-weather.yaml`. Settings you can tune are in the `substitutions:` block at the top of each file.

Awning: first-time setup

1. Flash with the motor wires **disconnected**. Check with a meter that M+ and M- stay at 0 V through a reboot. If either goes to 12 V at boot, check the 10k pull-downs on RPWM and LPWM.
2. Connect the motor. Try the wall rocker both ways, then the Home Assistant buttons.
3. Stop the awning closed, at about 25, 50 and 75%, and fully open. At each stop, note **Arm Tilt** and the measured extension, and enter the pairs under `calibrate_linear` (smallest tilt first). Then set `position_calibrated: "true"` and flash again. Until you do, it stops at full open only when the arm starts to rise again (the fabric beginning to roll back up upside down), so during calibration stop it yourself just before the end.
4. Time a full open and a full close and set `max_travel_ms` about 3 seconds above the slower one. Watch **Motor Current** during a normal run and set `motor_stall_amps` a few amps above it.
5. Watch **Wind Shake** on a breezy day, set `wind_shake_threshold` above normal flapping, then turn on **Wind Auto-Retract**.

What shows up in Home Assistant

- Awning cover with position, plus Status ("Open", "Stopped at 40%", "Problem: stuck at 35%", "Wind: auto-retracting")
- Arm Tilt, Wind Shake, Wind Detected, Wind Auto-Retract, Motor Current, Last Open/Close Time
- Battery Voltage, Current, Power, State of Charge, plus a "Battery Set Full" button
- Tank levels, Water Pump and Water Heater switches, Water Main and Tank Dump valves, Leak Lockout

> **Safety:** this project switches 12 V loads, a motor and a propane appliance's ignition signal. Fuse every feed at the source, use wire sized for the current, test with loads disconnected first, and never bypass factory safety controls. You build and use it at your own risk.